

DBMS WEEKLY TASK 2

Identifying Data Requirements for an Online Retail System

About Primary Key

A Primary Key (PK) is a unique identifier assigned to every record in a database table. It helps the database distinguish one record from another and ensures that duplicate records are not created. A primary key cannot contain NULL values and is commonly used to establish relationships between multiple tables.

Example:

Every customer is assigned a unique Customer_ID such as CUS001. Even if two customers have the same name, their Customer_ID will always be different.

Main Data Entities

1. Customer

Stores personal details of customers using the shopping platform.

Attributes

Customer_ID (Primary Key)

Customer_Name

Email

Phone_Number

Password

Address

Date_of_Birth

2. Product

Contains information about products sold on the website.

Attributes

Product_ID (Primary Key)

Product_Name

Brand_Name

Price

Quantity_Available

Category_ID

Supplier_ID

3. Category

Used to organize products into different groups.

Attributes

Category_ID (Primary Key)

Category_Name

Description

4. Seller

Stores details of registered sellers.

Attributes

Seller_ID (Primary Key)

Seller_Name

Business_Name

Contact_Number

Email

Address

5. Inventory

Maintains stock information.

Attributes

Inventory_ID (Primary Key)

Product_ID

Stock_Quantity

Last_Updated

6. Orders

Stores purchase details.

Attributes

Order_ID (Primary Key)

Customer_ID

Order_Date

Order_Status

Total_Price

7. Payment

Stores payment records.

Attributes

Payment_ID (Primary Key)

Order_ID

Payment_Mode

Amount

Payment_Status

8. Delivery

Tracks the shipment process.

Attributes

Delivery_ID (Primary Key)

Order_ID

Courier_Name

Tracking_ID

Delivery_Status

Additional Data Requirements

Search History

Stores recently searched products.

Attributes

Search_ID

Customer_ID

Search_Text

Search_Date

Product Filters

Used to display products based on customer preferences.

Filter Options

Category

Brand

Price

Customer Rating

Discount

Availability

Color

Material

Sorting Options

Helps customers arrange products.

Sort By

Best Match

Latest Products

Lowest Price

Highest Price

Most Popular

Gender Category

Useful for clothing and fashion products.

Types

Men

Women

Kids

Unisex

Coupons

Stores promotional offers available to customers.

Attributes

Coupon_ID

Coupon_Code

Discount_Value

Expiry_Date

Return & Refund

Maintains information about returned products.

Attributes

Return_ID

Order_ID

Return_Reason

Refund_Status

Return_Date

Conclusion